

Nail Kocabay

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SUMMARY

- Computer Engineering student at Marmara University with all fourth-year coursework completed and only a mandatory **18-day internship** remaining for graduation. Experienced in **AI/ML, Backend, Full-Stack Development, and Data Analysis**, building AI systems, data pipelines, and backend applications.

EXPERIENCE

AI Engineering Intern

September 2025 – May 2026

Payantis Finansal Teknoloji A.Ş., İstanbul

- Developed end-to-end **machine learning pipelines** for fraud detection across **16M+ transactions**, covering preprocessing, feature engineering, card profiling, and fraud reasoning.
- Engineered **13 behavioral features** capturing transaction velocity, spending patterns, temporal behavior, and geographic/MCC changes.
- Applied **MiniBatchKMeans** and **GPU-accelerated RAPIDS cuML KMeans** for card behavioral profiling and fraud analysis.
- Trained an **XGBoost fraud detection model** using transaction, card, merchant, and temporal features, achieving **99.56% fraud recall** and detecting **4,775 of 4,796 fraudulent transactions** on a **5M+ transaction test set**.
- Built a **multi-agent AI fraud reasoning system** and optimized large-scale processing using **Pandas, Dask, parallel processing, and GPU acceleration**.

Engineering Intern

July 2025 – September 2025

JMW Jant Sanayi A.Ş., Aydın

- Worked on **Windows Server administration, system configuration, network security, and enterprise IT infrastructure**.
- Assisted with IT system monitoring, server management, troubleshooting, and network-related issue investigation.

Intern

August 2024 – September 2024

FLO Mağazacılık A.Ş., İstanbul

- Developed responsive web page components using **HTML5 and CSS3**, applying semantic HTML and CSS layout techniques.
- Recreated company web interface components while gaining practical experience in **frontend development and responsive design**.

PROJECTS

Multi-Agent Fraud Reasoning System

- Designed a **multi-agent AI system** for fraud analysis using specialized agents for data analysis, financial reasoning, legal analysis, and final decision making.
- Integrated transaction history, behavioral features, and **card-level clustering** using LLMs and structured prompts.
- Achieved **71.43% precision, 100% recall, 83.33% binary F1, and 79.17% macro F1** in evaluation.

Full-Stack E-Commerce Platform

- Developed a full-stack e-commerce application using **ASP.NET Core 8, React 18, SQL Server, and Entity Framework Core**.
- Applied **Clean Architecture, CQRS, Domain-Driven Design, MediatR, and FluentValidation**.
- Implemented **JWT authentication and RBAC** and containerized the application using **Docker and Docker Compose**.

Spotify Music Recommendation Platform

- Developed a content-based recommendation platform using **Python, FastAPI, scikit-learn, Pandas, and NumPy**.
- Cleaned a **114K+ track dataset**, reducing it to **89,740 usable records** through duplicate and missing-value handling.
- Implemented a **KNN recommendation engine** using cosine distance, feature scaling, and **10 audio features**.

HTTP Web Server Project

- Built an HTTP server from scratch using **Python socket programming**, supporting multiple client connections through threading.
- Implemented **static file serving, MIME detection, JSON APIs, logging, and HTTP error handling**, with Docker containerization.

TECHNICAL SKILLS

Languages: Python, C#, Java, C, SQL, JavaScript, TypeScript, HTML5, CSS3

AI/ML: Machine Learning, Fraud Detection, Anomaly Detection, Feature Engineering, Clustering, KNN, Recommendation Systems, Multi-Agent AI

Frameworks: ASP.NET Core, React, Next.js, FastAPI, scikit-learn, Pandas, NumPy, Entity Framework Core

Backend: REST APIs, Clean Architecture, CQRS, DDD, MediatR, JWT, RBAC, Socket Programming

Databases/Tools: SQL Server, PostgreSQL, Git, GitHub, Docker, Docker Compose, Azure, GCP

EDUCATION

Marmara University

Computer Engineering, 83.8 / 100

Istanbul, Türkiye

2022 – 2026

LANGUAGE

Turkish: Native

English: B2